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RANDOMNESS, NONSENSE, AND THE SCIENTIFIC INTELLECTUAL

On extending the Monte Carlo generator to produce artificial thinking and compare it with rigorous non-random constructs. The science wars enter the business world. Why the aesthete in me loves to be fooled by randomness.

Randomness and the Verb

Our Monte Carlo engine can take us into more literary territory. Increasingly, a distinction is being made between the scientific intellectual and the literary intellectual – culminating with what is called the "science wars", plotting factions of literate non-scientists against no less literate scientists. The distinction between the two approaches originated in Vienna in the 1930s, with a collection of physicists who decided that the large gains in science were becoming significant enough to make claims on the field known to belong to the humanities. In their view, literary thinking could conceal plenty of well-sounding nonsense. They wanted to strip thinking from rhetoric (except in literature and poetry where it properly belonged).

The way they introduced rigor into intellectual life is by declaring that a statement could fall only into two categories: *deductive*, like "2 + 2 = 4", i.e., incontrovertibly flowing from a precisely defined axiomatic framework (here the rules of arithmetic), or *inductive*, i.e., verifiable in some manner (experience, statistics, etc.), like "it rains in Spain" or "New Yorkers are generally rude". Anything else was plain unadulterated hogwash (music could be a far better replacement to metaphysics). Needless to say that inductive statements may turn out to be difficult, even impossible, to verify, as we will see with the black swan problem – and empiricism can be worse than any other form of hogwash when it gives someone confidence (it will take me a few chapters to drill the point). However, it was a good start to make intellectuals responsible for providing some form of evidence for their statements. This Vienna Circle was at the origin of the development of the ideas of Wittgenstein, Popper, Carnap, and flocks of others. Whatever merit their original ideas may have, the impact on both philosophy and the practice of science has been significant. Some of their impact on non-philosophical intellectual life is starting to develop, albeit considerably more slowly.

One conceivable way to discriminate between a scientific intellectual and a literary intellectual is by considering that a scientific intellectual can usually recognize the writing of another but that the literary intellectual would not be able to tell the difference between lines jotted down by a scientist and those by a glib non-scientist. This is even more apparent when the literary intellectual starts using scientific buzzwords, like "uncertainty principle", "Gödel's theorem", "parallel universe", or "relativity" either out of context or, as often, in exact opposition to the scientific meaning. I suggest reading the hilarious *Fashionable Nonsense* by Alan Sokal for an illustration of such practice (I was laughing so loudly and so frequently while reading it on a plane that other passengers kept whispering things about me). By dumping the kitchen sink of scientific references in a paper, one can make another literary intellectual believe that one's material has the stamp of science. Clearly, to a scientist, science lies in the rigor of the inference, not in random references to such grandiose concepts as general relativity or quantum indeterminacy. Such rigor can be spelled out in plain English. Science is method and rigor; it can be identified in the simplest of prose writing. For instance, what